

# Essential ML Research Papers for TrustScan

## 1. Forgery & Image Manipulation Detection

- **"A Picture is Worth a Thousand Lies: Photo Forensics" (H. Farid, 2009)**

A foundational guide to digital forensics explaining how to detect cloning, resampling, and lighting inconsistencies.

- **"Digital Image Forgery Detection: A Survey" (S. Walia and K. Kumar, 2019)**

A comprehensive look at image forensics, including traditional methods and modern deep learning approaches.

## 2. Document Intelligence & OCR

- **"LayoutLM: Pre-training of Text and Layout for Document Image Understanding" (Xu et al., 2020)**

A model that learns both text and its location, ideal for structured document extraction.

- **"Donut: Document Understanding Transformer without OCR" (Kim et al., 2022)**

Explores OCR-free document understanding, directly mapping images to structured data.

## 3. Fraud Detection & Imbalanced Learning

- **"SMOTE: Synthetic Minority Over-sampling Technique" (Chawla et al., 2002)**

The standard method for dealing with imbalanced datasets by generating synthetic minority samples.

- **"Random Forests" (Leo Breiman, 2001)**

Explaining the tree-based ensemble method that handles non-linear relationships better than basic models.

## 4. General ML Stability & Reliability

- **"Attention Is All You Need" (Vaswani et al., 2017)**

The foundational paper for Transformers, introducing the attention mechanism used in modern LLMs.

- **"Hidden Technical Debt in Machine Learning Systems" (Sculley et al., Google, 2015)**

Focuses on the engineering challenges of maintaining ML systems in production.